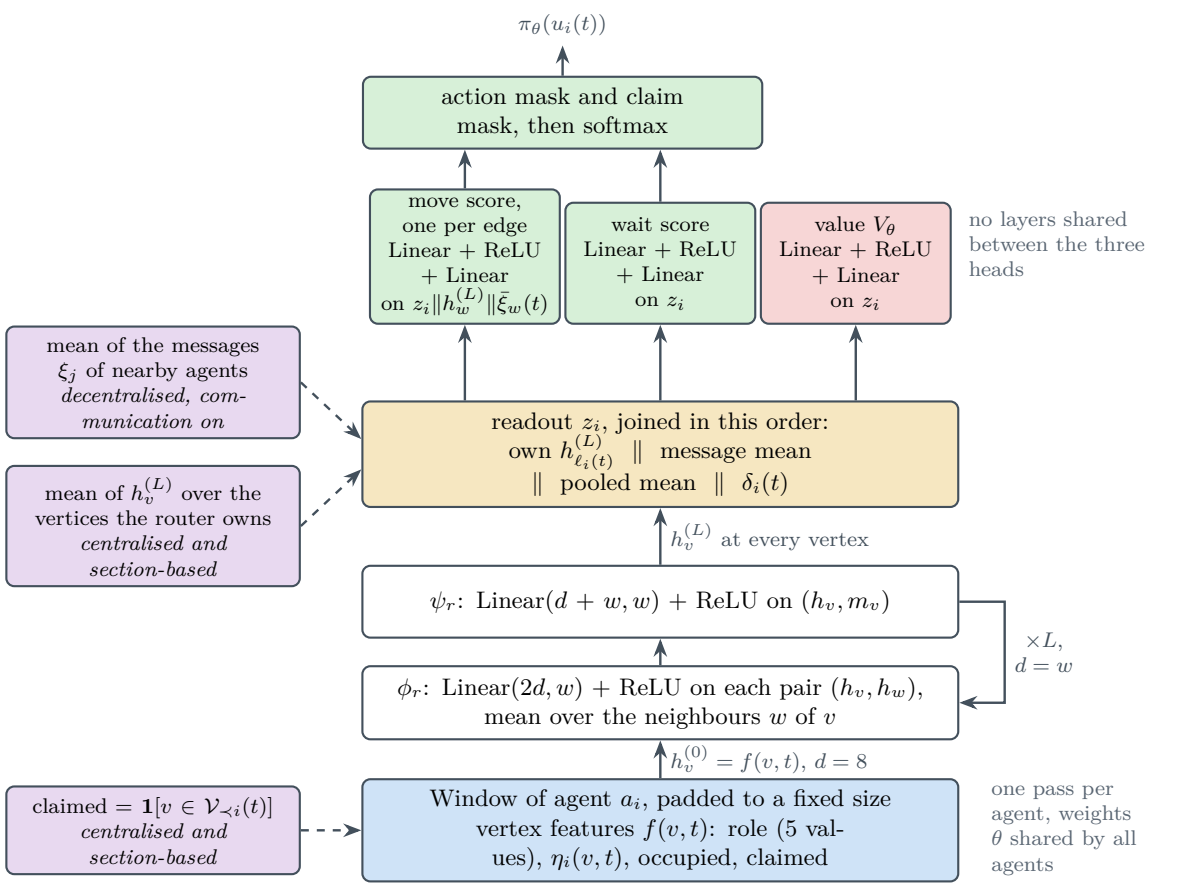

$\pi_\theta(u_i(t))$
action mask and claim mask, then softmax
move score, one per edge Linear + ReLU + Linear on $z_i \| h_w^{(L)} \| \bar{\xi}_w(t)$
wait score Linear + ReLU + Linear on z_i
value V_θ Linear + ReLU + Linear on z_i
no layers shared between the three heads
mean of the messages ξ_j of nearby agents *decentralised, communication on*
mean of $h_v^{(L)}$ over the vertices the router owns *centralised and section-based*
readout z_i, joined in this order: own $h_{\ell_i(t)}^{(L)}$ ‖ message mean ‖ pooled mean ‖ $\delta_i(t)$
$h_v^{(L)}$ at every vertex
ψ_r: Linear$(d + w, w)$ + ReLU on (h_v, m_v)
ϕ_r: Linear$(2d, w)$ + ReLU on each pair (h_v, h_w), mean over the neighbours w of v
$\times L$, $d = w$
$h_v^{(0)} = f(v,t)$, $d = 8$
claimed $= \mathbf{1}[v \in \mathcal{V}_{\prec i}(t)]$ *centralised and section-based*
Window of agent a_i, padded to a fixed size vertex features $f(v,t)$: role (5 values), $\eta_i(v,t)$, occupied, claimed
one pass per agent, weights θ shared by all agents